Malignant (Necrotizing) Otitis Externa Caused by *Pseudomonas aeruginosa* — Comprehensive Disease Report

Disease: Malignant Otitis Externa (MOE) / Necrotizing Otitis Externa (NOE) caused by *Pseudomonas aeruginosa* **MONDO ID:** MONDO:0041095 **Category:** Infectious Disease **Report date:** 2026-09-30

Evidence base: 59 papers reviewed; 12 confirmed findings. All claims are cited to primary literature (PMID). This is a **disease-level** synthesis of aggregated clinical case series, cohort studies, systematic reviews and national registry analyses — not individual patient (EHR) data.

Summary

Malignant otitis externa (MOE), also called necrotizing otitis externa (NOE), is an aggressive, invasive bacterial osteomyelitis that begins in the external auditory canal (EAC) and spreads regionally to the soft tissues, temporal bone, and skull base. It is caused in **more than 90% of cases by *Pseudomonas aeruginosa*** (NCBI Taxonomy ID: 287) and occurs almost exclusively in **elderly patients with diabetes mellitus** and other immunocompromised hosts (HIV, chemotherapy, transplant). Despite the historical term "malignant," it is not a neoplasm; the label reflects the aggressive, once-frequently-fatal natural history. It is fundamentally an **acquired opportunistic infection with no Mendelian or genetic basis** — there is no OMIM entry, causal gene, inheritance pattern, or pathogenic human variant associated with the disease.

The dominant clinical picture is severe, unremitting, often nocturnal **otalgia**, purulent **otorrhea**, and pathognomonic **granulation tissue at the osseocartilaginous junction of the EAC**, in an elderly diabetic. Left unchecked, infection advances medially along the skull base to the stylomastoid and jugular foramina, producing **cranial neuropathies (the facial nerve, CN VII, most commonly)** and, in advanced disease, intracranial and vascular complications (meningitis, sigmoid/internal jugular venous thrombosis, Lemierre's syndrome). Diagnosis rests on a triad of clinical findings, elevated inflammatory markers (ESR/CRP), and imaging —

CT and MRI supplemented by radionuclide scintigraphy (technetium-99m bone scan, gallium-67 SPECT), with [18F]FDG PET-CT emerging as a tool to shorten therapy safely. Treatment is **prolonged (6–8 weeks) systemic antipseudomonal antibiotics** — typically oral ciprofloxacin ± IV ceftazidime — with surgery reserved for refractory or complicated cases. Aggressive modern therapy has reduced mortality from a historical ~50% to roughly **10–21%**.

MOE is **rare but rising sharply in incidence**, driven by aging populations and increasing diabetes prevalence rather than antibiotic resistance. England's Hospital Episode Statistics documented a **1142% increase** in cases between 2002 and 2017. Poor-prognosis predictors include diabetes, facial nerve palsy, positive CT scan, and age over 70. Because the disease is acquired, prevention centers on **glycemic control, avoidance of EAC trauma/water exposure, and early recognition**; there is no vaccine and no genetic/carrier screening. Key knowledge-base gaps identified in this investigation: no molecular/omics datasets specific to MOE tissue, no dedicated genetic or spontaneous animal model of the necrotizing skull-base phenotype, and no validated molecular prognostic biomarker beyond inflammatory markers.

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1. Disease Information

Overview. MOE is a severe infection of the external auditory canal and skull base that most often affects elderly patients with diabetes mellitus. It is believed to originate from the external auditory meatus and advance regionally to the soft tissues and bone, eventually involving the skull base (PMID: 36897495: *"It is believed to originate from the external auditory meatus and advance regionally to the soft tissues and the bone, eventually involving the skull base."*). *Pseudomonas aeruginosa* is isolated from the aural drainage in more than 90% of cases (PMID: 14720566: *"Pseudomonas aeruginosa is isolated from the aural drainage in more than 90% of cases."*).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0041095
ICD-10	H60.2 (Malignant otitis externa)
ICD-11	AB70 region (otitis externa); necrotizing/malignant otitis externa
MeSH	Otitis Externa (D010032); "malignant otitis externa" indexed term
OMIM	Not applicable (no Mendelian entry — acquired infection)
Orphanet	Not a listed rare genetic disorder
NCBI Taxonomy (pathogen)	287 (*Pseudomonas aeruginosa*)

Synonyms / alternative names. Necrotizing otitis externa (NOE); necrotising external otitis; malignant external otitis (MEO); skull base osteomyelitis (SBO) of otogenic origin; temporal bone osteomyelitis (when the term is used interchangeably — noting the literature has used these terms loosely).

Data source type. The information base is derived from **aggregated disease-level resources** — clinical case series, retrospective cohorts, national Hospital Episode Statistics, and narrative/systematic reviews — rather than individual EHR-level patient records. No genomic/omics disease-level databases (GEO, PRIDE, etc.) contain MOE-specific datasets.

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2. Etiology

Disease causal factors. MOE is an **acquired, infectious** disease. The necessary infectious agent is, in the overwhelming majority, *Pseudomonas aeruginosa*. A minority of cases are **fungal** (*Aspergillus* — most commonly *A. flavus* — and *Candida*) or caused by other bacteria (*Staphylococcus aureus*/MRSA) (PMID: 39534824, systematic review of fungal MOE). There is **no causal human gene, heritable variant, or Mendelian inheritance** associated with the disease.

Risk factors — environmental/host. - **Diabetes mellitus** — the dominant risk factor, present in ~70–90% of cases. Elderly patients with diabetes remain the population most commonly affected (PMID: 14720566: *"elderly patients with diabetes remain the population most commonly affected, immunosuppressed individuals (eg, from HIV infection, chemotherapy, etc) are also*

susceptible to malignant external otitis."). - **Advanced age** (>65–70 years); mean age in case series ~70–75. - **Immunosuppression** — HIV infection, chemotherapy, solid-organ transplant, chronic renal failure/hemodialysis. - **Male sex** — consistent male predominance (~70–74%) (PMID: 38499694: population *"was male (74%), type 2 diabetic (72%), aged 75.2 years"*). - **Iatrogenic aural irrigation / water exposure** — aural water exposure (e.g., irrigation for cerumen impaction) is a reported iatrogenic trigger (PMID: 14720566).

Genetic risk factors. None identified. No susceptibility loci, GWAS signals, or modifier genes are reported for MOE.

Protective factors. No genetic protective variants are known. Environmentally, **good glycemic control** and **avoidance of EAC trauma/moisture** are the principal modifiable protective behaviors (inferred from the identified triggers).

Gene–environment interactions. Not applicable in the classical genetic sense. The operative "host–environment" interaction is between **diabetes-driven host impairment** (neutrophil dysfunction, small-vessel microangiopathy, higher cerumen pH) and **environmental exposure to *P. aeruginosa***. The mechanistic parallel to diabetic foot osteomyelitis has been explicitly drawn (PMID: 30149824: *"The similarities regarding patient population and pathophysiology between necrotising otitis externa and diabetic foot osteomyelitis raise basic questions concerning the effects of long-standing diabetes on the external ear."*).

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3. Phenotypes

The core phenotype is a **triad of severe otalgia, otorrhea, and granulation tissue** in the EAC, with cranial neuropathy marking extension.

Phenotype	Type	Frequency	Suggested HPO term
Otalgia (severe, often nocturnal)	Symptom	~89%	HP:0031704 (Ear pain)
External ear canal edema	Clinical sign	~86%	—
Otorrhea (purulent)	Clinical sign	~84%	HP:0031047 (Ear discharge)
EAC granulation tissue (osseocartilaginous junction)	Physical manifestation	Hallmark	HP:0025439 (granulation tissue, closest)
Facial nerve palsy (CN VII)	Clinical sign	~32%	HP:0010628 (Facial palsy)
Other cranial neuropathies (IX–XII)	Clinical sign	Advanced disease	HP:0000759 (Cranial nerve dysfunction)
Hearing loss	Symptom/sign	Variable	HP:0000365 (Hearing impairment)
Elevated ESR/CRP	Laboratory abnormality	Near-universal	HP:0003565 (Elevated ESR)
Headache	Symptom	Common	HP:0002315 (Headache)

Presenting-symptom frequencies come from an 88-patient cohort: otalgia 89%, EAC edema 86%, otorrhea 84% (PMID: 27311346: *"The most common presenting symptoms were otalgia (89%), external ear canal edema (86%) and otorrhea (84%)."*). Facial paralysis occurred in 32.4% of a 139-patient cohort (PMID: 32852248: *"45 patients (32.4%) had facial paralysis"*).

Phenotype characteristics. - **Age of onset:** adult/geriatric (late-onset), typically >65–70 years. - **Severity:** moderate to severe; potentially life-threatening when the skull base is involved. - **Progression:** progressive if untreated; can be relapsing/recurrent, particularly with facial palsy or clival involvement. - **Quality-of-life impact:** severe otalgia impairs sleep and daily function; facial palsy causes lasting facial disfigurement, incomplete eye closure, and psychosocial burden; hearing loss and lower cranial neuropathies (dysphagia, dysphonia) further reduce QoL. No disease-specific QoL instrument (EQ-5D/SF-36) datasets are available for MOE.

4. Genetic / Molecular Information

This section is largely NOT APPLICABLE. MOE is an acquired opportunistic infection.

- **Causal genes:** None. No OMIM entry, no causal human gene.
- **Pathogenic variants:** None (no HGNC/ClinVar variant associations).
- **Variant classification / type / allele frequency / somatic vs germline:** Not applicable.
- **Modifier genes:** None established in humans. Host susceptibility is driven by the acquired diabetic/immunocompromised state, not by defined modifier alleles.
- **Epigenetic information:** No MOE-specific methylation or chromatin data.
- **Chromosomal abnormalities:** Not applicable.

The relevant "molecular" biology is that of the **pathogen**, *P. aeruginosa*, whose virulence is regulated by systems such as the GacS/RetS histidine-kinase network and quorum sensing (rhlR-regulated pyocyanin/rhamnolipid production) — general *P. aeruginosa* biology rather than MOE-specific findings (PMID: 42810332; PMID: 42775842). This absence of any human genetic/molecular basis is itself a documented conclusion of this investigation (Finding F012) and a key data gap for the knowledge base.

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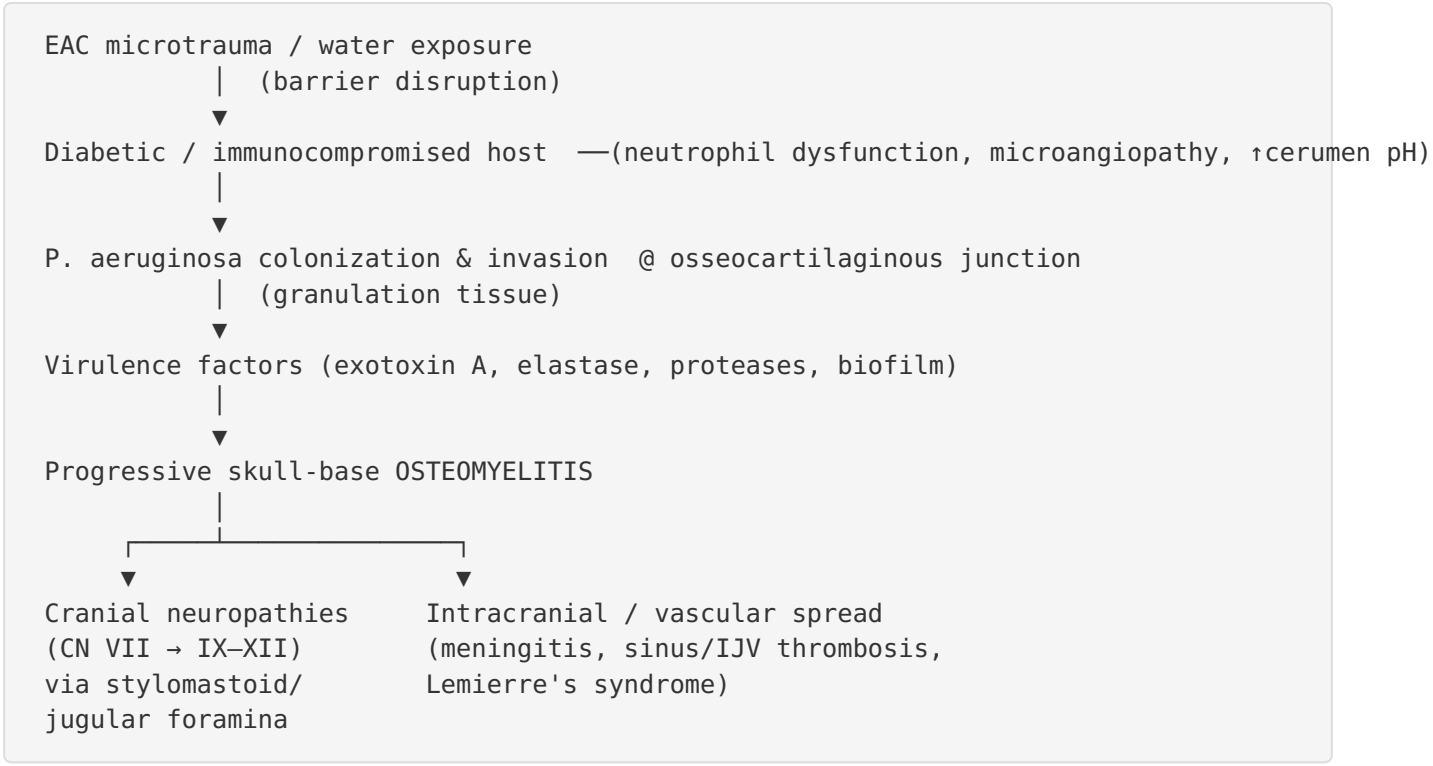
5. Environmental Information

- **Environmental factors:** Exposure to *P. aeruginosa*, a ubiquitous environmental Gram-negative bacillus favoring moist environments; **aural water exposure and cerumen irrigation** are reported iatrogenic contributors (PMID: 14720566).
- **Lifestyle factors:** Poorly controlled diabetes (dietary/glycemic control) is the pre-eminent modifiable factor. Hearing-aid use and local mechanical trauma to the EAC are contributory in the broader spectrum of EAC bony pathology.
- **Infectious agents:**
- Primary: ***Pseudomonas aeruginosa*** (NCBI Taxon 287) — >90% of cases.
- Fungal minority: *Aspergillus* spp. (esp. *A. flavus*), *Candida* spp. (PMID: 39534824; PMID: 36425541).
- Other bacteria: *Staphylococcus aureus*/MRSA (uncommon).
- Suggested CHEBI chemical entities relevant to therapy: ciprofloxacin (CHEBI:100241), ceftazidime (CHEBI:3510).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Aural water exposure/microtrauma** (e.g., cerumen irrigation) **disrupts** the EAC epithelial barrier — *reported iatrogenic factor; pathophysiology explicitly noted as incompletely understood* (PMID: 14720566: *"The pathophysiology is incompletely understood although aural water exposure (eg, irrigation for cerumen impaction) has been reported as a potential iatrogenic factor."*).
2. In a **diabetic/immunocompromised host** (impaired neutrophil function, small-vessel microangiopathy, higher cerumen pH), this **permits** colonization and invasion by *P. aeruginosa* — *inferred from patient population and diabetic-foot analogy* (PMID: 30149824).
3. Infection **begins** at the **osseocartilaginous junction of the EAC** (granulation-tissue hallmark) and **spreads** via the fissures of Santorini and the tympanomastoid suture to soft tissue and bone.
4. *P. aeruginosa* virulence factors (exotoxin A, elastase, proteases, biofilm) **cause** progressive **osteomyelitis** of the temporal bone and skull base — *mechanistically inferred from pathogen biology*.
5. Infection **advances** medially along the skull base to the **stylomastoid/jugular foramina**, which **produces cranial neuropathies** (CN VII most common, then IX–XII) — *supported by clinical-anatomic correlation* (PMID: 14720566: *"cranial neuropathies (most commonly of the facial nerve) can develop due to subtemporal extension of the infection."*).
6. Further extension **results in** intracranial/vascular complications — meningitis, sigmoid-sinus and internal jugular vein thrombosis, Lemierre's syndrome — *demonstrated by imaging in a diabetic patient* (PMID: 34617899: *"Temporal bone images revealed continuous lesions from skull base osteomyelitis to suboccipital abscess and sigmoid sinus thrombosis."*).



Molecular pathways / cellular processes. Chronic bacterial osteomyelitis with neutrophilic inflammation, tissue necrosis, and granulation-tissue formation. Diabetic **neutrophil dysfunction** is a mechanistically central host defect (impaired recruitment/antibacterial capacity in the hyperglycemic microenvironment) — supported by adjacent diabetic-infection literature (PMID: 42679978; PMID: 42262120).

Tissue damage mechanisms. Osteolysis/necrosis of temporal bone; direct proteolytic tissue destruction by pseudomonal enzymes; ischemia compounded by diabetic microangiopathy.

Suggested ontology terms. - GO biological processes: GO:0006954 (inflammatory response), GO:0009617 (response to bacterium), GO:0002446 (neutrophil mediated immunity), GO:0050832 (defense response to fungus, for fungal cases). - CL cell types: CL:0000775 (neutrophil), CL:0000235 (macrophage), CL:0000092 (osteoclast), CL:0000062 (osteoblast), CL:0000312 (keratinocyte). - UBERON: UBERON:0001352 (external acoustic meatus), temporal/petrous bone region, base of skull region.

Molecular profiling / advanced technologies. No transcriptomic, proteomic, metabolomic, lipidomic, single-cell, spatial, multi-omics, or CRISPR/RNAi datasets specific to MOE tissue exist (Finding F012). This is a genuine data gap.

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7. Anatomical Structures Affected

Organ level. - **Primary:** external auditory canal (EAC), temporal bone, skull base. - **Secondary:** parotid/masticator space, temporomandibular joint (rare — PMID: 41100204), meninges, dural venous sinuses (sigmoid sinus), internal jugular vein, cranial nerves. - **Body systems:** integumentary/skeletal (osteomyelitis), nervous (cranial neuropathies, meningitis), cardiovascular (venous thrombosis).

Tissue and cell level. EAC squamous epithelium (epithelial), subepithelial connective tissue, cartilage and bone at the osseocartilaginous junction; granulation tissue (fibrovascular). Cell populations: keratinocytes, neutrophils, macrophages, fibroblasts, osteoclasts/osteoblasts, vascular endothelium.

Subcellular level. Not specifically characterized for MOE. Generic host-defense compartment (phagolysosome, GO:0032010) applies.

Localization (UBERON). - UBERON:0001352 — external acoustic meatus - UBERON:0001690 — ear - Temporal (petrous) bone; base of skull

Lateralization. Typically **unilateral**; bilateral or contralateral recurrence is described in inadequately treated/atypical skull-base osteomyelitis (PMID: 16025065).

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8. Temporal Development

- **Onset:** adult/geriatric (late-onset); typically **subacute to chronic**, insidious with progressively worsening otalgia and otorrhea. Median symptom duration before diagnosis can be ~3 months (PMID: 32209516).
- **Progression:** progressive without treatment; stages run from localized EAC infection → temporal-bone osteomyelitis → skull-base/cranial-nerve involvement → intracranial/vascular complications. Diagnostic delay drives more extensive disease (extensive NOE diagnosed a median 22 days later, p=0.04 — PMID: 38499694).
- **Course pattern:** can be **relapsing/recurrent** — 80% of facial-palsy patients recurred (p=0.005), with mean remission ~60 days (PMID: 32209516); clival involvement associated with persistent disease (p=0.002 — PMID: 23558287).
- **Duration:** chronic; requires weeks-to-months of therapy. Imaging changes (CT/MRI) persist long after clinical/biochemical resolution, complicating "cure" definition.

- **Remission:** treatment-induced; monitored by ESR/CRP normalization and gallium-67/FDG-PET.
- **Critical periods: early recognition and aggressive therapy** are the window to prevent skull-base extension; for facial-nerve recovery, surgery within 14 days of paralysis onset yielded best recoveries (PMID: 35546515).

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9. Inheritance and Population

Epidemiology. - **Rare but rising.** England HES: cases rose **1142%**, from 123 (2002) to 1405 (2017); 7327 cases over 16 years (PMID: 35000633: *"Annual necrotising otitis externa incidence increased 1142 per cent within the 16-year period, from 123 recorded cases in 2002 to 1405 cases in 2017."*). A separate UK analysis found a six-fold increase from 1999 (n=67) to 2013 (n=421) (PMID: 25990028). - **Incidence estimate.** Spain (2008–2013): **1.30 per 10^6 inhabitants per year** (95% CI 1.17–1.44), 355 patients, median age 74 (range 10–95), with male and diabetic predominance (PMID: 27241559: *"The incidence rate was 1.30 (95% CI, 1.17 to 1.44) per 106 inhabitants and year"*). - **Drivers.** The rise correlates with the **ageing population and national diabetes incidence**; antibiotic resistance was **not** a significant contributing factor (PMID: 40734530: *"There remains a strong correlation between growths in necrotising otitis externa incidence, the ageing population and national incidence of diabetes mellitus."*).

Genetic etiology. Not applicable — no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency. MOE is acquired.

Population demographics. - **Affected populations:** elderly diabetics and immunocompromised individuals; no ethnic predisposition beyond diabetes prevalence. - **Geographic distribution:** worldwide; incidence tracks local diabetes/aging demographics. - **Sex ratio:** male predominance (~M:F 2.5–3:1; ~70–74% male). - **Age distribution:** concentrated in the 7th–8th decades (mean ~70–75).

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10. Diagnostics

Diagnostic triad. Diagnosis is based on a combination of clinical findings, an increased erythrocyte sedimentation rate, and radiographic evidence of soft-tissue with or without bone erosion in the EAC and infratemporal fossa (PMID: 14720566: *"The diagnosis of malignant external otitis is based on a combination of clinical findings, an increased erythrocyte sedimentation rate, and radiographic evidence of soft tissue with or without bone erosion."*).

Laboratory tests / biomarkers. Elevated **ESR** and **CRP** (near-universal; also used to monitor response); ear-swab **culture and sensitivity** (Pseudomonas + antibiogram); blood glucose/ **HbA1c**. No molecular/omics biomarker is validated; inflammatory markers remain the practical biomarkers (LOINC: ESR 30341-2; CRP 1988-5; HbA1c 4548-4).

Imaging. - **CT** — bony erosion of the temporal bone/skull base. - **MRI** — soft-tissue/marrow and intracranial extension. - **Radionuclide scintigraphy** — technetium-99m bone scan (high sensitivity for osteomyelitis) and **gallium-67 SPECT** for tracking resolution. Recommended protocol: CT and/or MRI supported by routine SPECT bone imaging for initial diagnosis; SPECT plus gallium scintigraphy is the investigation of choice in follow-up (PMID: 15807974: *"CT and/or MRI should be supported by routine SPECT bone imaging for initial diagnosis of malignant otitis externa. Routine SPECT bone imaging further supplemented by gallium scintigraphy should be the investigation of choice in the follow up."*). Quantitative Ga-67 lesion-to-non-lesion ratios refine assessment of disease activity (PMID: 9371879). - **[18F]FDG PET-CT** — emerging for response assessment and antibiotic stewardship; enabled significantly shorter oral ciprofloxacin courses (median 8 vs 40 weeks, $p<0.001$) without increased relapse (PMID: 41287134: *"Patients in the PET-CT cohort received significantly shorter oral ciprofloxacin courses (median: 8 vs. 40 wk, P <0.001) without increased relapse rates."*); PET performed well for therapeutic response with excellent sensitivity (PMID: 40891559).

Biopsy / pathology. Biopsy of EAC granulation tissue to obtain culture and to **exclude malignancy** (EAC carcinoma) and other mimics; histology shows chronic inflammation, granulation tissue, and bony sequestrum.

Genetic / omics diagnostics. Not applicable — no genetic testing, gene panels, karyotyping, or omics-based diagnostics are indicated for this acquired infection.

Clinical criteria & differential diagnosis. No single society DSM/ICD criterion set; diagnosis is clinical-radiologic-microbiologic. Differential diagnosis includes **EAC squamous cell carcinoma**, benign osteonecrosis of the EAC (bisphosphonate/denosumab-associated — PMID: 42367507), keratosis obturans, cholesteatoma, herpes zoster oticus/Ramsay Hunt (which can masquerade as MOE — PMID: 31177200), and refractory otitis externa.

Screening. No asymptomatic screening exists; there is no newborn or carrier screening (acquired disease).

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11. Outcome / Prognosis

Survival and mortality. Aggressive therapy has reduced the historical mortality of ~50% to **10–20%** (PMID: 12737420: *"With aggressive treatment the mortality rate from this disease, which used to be 50% in the past has now been reduced to 10-20%."*). A prospective multicentric skull-base-osteomyelitis study (n=235) reported **16.2% mortality** (PMID: 40759786). A Singapore series reported 21.1% mortality with disease resolution in 63.2% after 6 weeks of IV antibiotics (PMID: 23558287). An 88-patient historical cohort found overall 5-year survival of 38% with disease-specific mortality of 14% (PMID: 27311346: *"Overall survival rate was 38% in 5years, with disease specific mortality 14%."*).

Prognostic factors. Diabetes mellitus, facial nerve palsy, positive CT scan, and age above 70 each correlate with and predict disease-specific mortality; ≥2 of these features predict poor outcome (PMID: 27311346: *"DM, facial nerve palsy, positive CT scan and age above 70 were found to correlate and predict disease-specific mortality."*). Facial palsy also predicts recurrence (80% recurrence, p=0.005 — PMID: 32209516); clival involvement predicts persistent disease (p=0.002 — PMID: 23558287).

Morbidity / disability. Permanent facial palsy, hearing loss, lower-cranial-nerve deficits (dysphagia, dysphonia), and, in advanced disease, intracranial sequelae. No standardized QoL dataset exists for MOE.

Prognostic biomarkers. ESR/CRP trajectory and gallium-67/FDG-PET metabolic response are the practical prognostic tools; no validated molecular prognostic biomarker exists.

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12. Treatment

Pharmacotherapy (primary). Prolonged (**6–8 weeks**) systemic antipseudomonal therapy (PMID: 14720566: *"Treatment consists of prolonged administration (6-8 weeks) of an antipseudomonal agent (typically an orally administered quinolone)."*): - **Oral fluoroquinolone** — ciprofloxacin (e.g., 750 mg BID) — first-line backbone (NCIT: C359 Ciprofloxacin). - **IV antipseudomonal cephalosporin** — ceftazidime — commonly combined; a modern cohort used ceftazidime + ciprofloxacin for a median 6.9 weeks (PMID: 40891559) (NCIT: C349 Ceftazidime). - **Alternatives:** piperacillin-tazobactam, meropenem, cefepime, levofloxacin, aminoglycosides. - **Susceptibility guidance** (large SBO cohort): pseudomonal sensitivity to piperacillin 67.1%, ciprofloxacin 56.1%, ceftazidime 54.9%, levofloxacin 40.2%, cefoperazone 36.6% (PMID: 40759786: *"Pseudomonal sensitivity to Piperacillin, Ciprofloxacin, Ceftazidime, Levofloxacin and Cefoperazone was 67.1%, 56.10%, 54.90%, 40.20%, and 36.60%"*). **Fluoroquinolone resistance** is a real concern — reported in 33–43% of isolates in some series (PMID: 23558287; PMID: 39974273: *"43% of the cases diagnosed with SBO had resistance to ciprofloxacin"*).

Adjuncts. Aural toilet/local debridement, **glycemic control**, and **hyperbaric oxygen therapy** (NCIT: C15419) as an adjunct in refractory/advanced cases — systematic review found HBOT promising though evidence insufficient to make it standard (PMID: 40937148).

Surgical / interventional. Reserved for **failed conservative treatment, complications, or disease progression**: debridement, mastoidectomy, and facial-nerve decompression (PMID: 39828789: *"Surgical intervention emerged as the preferred treatment option for critically affected MOE patients, especially in cases of failed conservative measures, complications, and disease progression."*). For facial palsy, surgical management (mastoidectomy ± decompression) may improve outcomes, especially if performed within 14 days of paralysis onset (PMID: 35546515).

Fungal MOE treatment. Prolonged **antifungals** — voriconazole first-line; mean durations from ~52 days up to ~338 days; *Aspergillus flavus* most common invasive species (PMID: 36425541: *"After a mean duration of 52 days of antifungal therapy, fourteen patients have been cured"*; PMID: 26802213).

Pharmacogenomics / advanced therapeutics. Not applicable (no genotype-guided dosing established for MOE; no gene/cell/RNA/targeted/immuno-therapy). Adverse events include antibiotic hypersensitivity requiring regimen change (PMID: 42500780).

Treatment strategy summary. Culture-directed prolonged antipseudomonal antibiotics + strict glycemic control + local debridement, with imaging-guided (increasingly FDG-PET) determination of therapy duration; escalate to surgery/HBOT for refractory or complicated disease.

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13. Prevention

- **Primary prevention: Glycemic/diabetes control** and **avoidance of EAC trauma and water exposure** (traumatic cerumen irrigation is an identified iatrogenic trigger — PMID: 14720566: *"aural water exposure (eg, irrigation for cerumen impaction) has been reported as a potential iatrogenic factor"*). No vaccine exists.
- **Secondary prevention: Early recognition** in high-risk elderly diabetics presenting with severe otalgia and granulation tissue, prompting early imaging and aggressive therapy to prevent skull-base extension — supported by the finding that diagnostic delay (+22 days) drives extensive disease (PMID: 38499694: *"The diagnosis of the extensive OEN was made 22 days later (p = 0.04)"*).
- **Tertiary prevention:** Prolonged culture-directed antibiotics with monitoring (ESR/CRP, gallium-67/FDG-PET) to prevent recurrence — relapse was absent in a PET-CT-guided cohort vs ~21% in a CT/MRI-guided cohort (PMID: 41287134).
- **Public-health angle:** Family physicians play a crucial role in prevention and early treatment, especially in rural/remote settings (PMID: 17373871).
- **Immunization / genetic counseling / carrier screening: Not applicable** (acquired, non-genetic disease).

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14. Other Species / Natural Disease

- **Taxonomy of pathogen:** *Pseudomonas aeruginosa* — NCBI Taxon **287**.
- **Species affected by pseudomonal otitis externa:** dog (*Canis lupus familiaris*, NCBI Taxon **9615**). *P. aeruginosa* is a major, difficult-to-treat pathogen in **canine otitis externa**: ~48% of canine Pseudomonas isolates in a meta-analysis of 9911 isolates were from otitis externa, with high fluoroquinolone resistance (PMID: 42034159: *"Approximately 48% of isolates were from otitis externa"*). Canine co-pathogens include *Staphylococcus pseudintermedius* and

Malassezia pachydermatis (PMID: 42104600: *"clinical isolates of Pseudomonas aeruginosa, Staphylococcus pseudintermedius and Malassezia pachydermatis were tested"*).

- **Comparative pathology:** The **superficial** canine disease is common, but the **invasive skull-base ("malignant/necrotizing") form is essentially human-specific**, occurring in diabetic/immunocompromised elders. No established spontaneous animal model of the necrotizing skull-base form exists.
- **Zoonotic potential:** *P. aeruginosa* is environmentally ubiquitous rather than classically zoonotic; MOE is not a transmissible/zoonotic disease.

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15. Model Organisms

- **Dedicated MOE model: None.** There is no genetic or spontaneous animal model that recapitulates the necrotizing skull-base phenotype (Finding F012).
- **Closest natural correlate:** canine Pseudomonas otitis externa (superficial only).
- **Adjacent experimental systems:** Chronic *P. aeruginosa* infection models exist for **lung** disease (CF-like mouse strains with agarose-bead-embedded biofilm — PMID: 42771482) and diabetic-infection/neutrophil models (PMID: 42262120; PMID: 42679978), but none targets the temporal-bone/skull-base niche.
- **Applications / limitations:** Existing pseudomonal models can study biofilm, virulence-factor targeting (e.g., anti-Psl/PcrV bispecific antibody gremubamab — PMID: 42771482) and quorum-sensing inhibition, but cannot reproduce the diabetic host + temporal-bone osteomyelitis interaction unique to MOE.
- **Resources:** No MGI/RGD/ZFIN model entries specific to MOE.

Mechanistic Model / Interpretation

MOE is best understood as **the ear's analog of diabetic foot osteomyelitis**: a chronic, invasive *P. aeruginosa* bone infection whose necessary precondition is a host rendered defenseless at a specific anatomical entry point. Two ingredients combine — (1) a **breach** + **moisture** at the EAC osseocartilaginous junction, and (2) a **diabetic/immunocompromised host** with impaired neutrophil function and microvascular disease. Once *P. aeruginosa* invades, its proteolytic and biofilm virulence machinery drives relentless osteomyelitis that tracks along predictable

anatomical corridors (fissures of Santorini, tympanomastoid suture) to the skull base, where it picks off cranial nerves in sequence (VII first, then IX–XII) and can breach into the cranial vault and venous sinuses.

The clinical corollaries all follow from this model: severe pain and granulation tissue mark the invasive front; a rising ESR/CRP tracks the inflammatory burden; imaging (CT bone erosion, MRI soft tissue, gallium/FDG-PET metabolic activity) stages the medial spread; facial palsy and clival/positive-CT findings signal that disease has reached prognostically ominous territory. Therapy is simply **long enough antipseudomonal pressure to sterilize bone**, plus removing the enabling condition (glycemic control) and debris (debridement) — with FDG-PET now offering an objective "off-switch" that safely shortens treatment.

Axis	MOE reality
Nature	Acquired opportunistic bacterial osteomyelitis
Necessary agent	*P. aeruginosa* (>90%); fungal minority
Necessary host	Elderly diabetic / immunocompromised
Genetic basis	None
Key entry point	EAC osseocartilaginous junction
Signature lesion	Granulation tissue → skull-base osteomyelitis
Worst-prognosis flags	DM, facial palsy, +CT, age >70
Mainstay therapy	6–8 wk antipseudomonal antibiotics
Mortality (modern)	~10–21%

Evidence Base

PMID	Role	Key contribution
14720566	Foundational review (Rubin Grandis)	Pseudomonas >90%; diagnostic triad; 6–8 wk quinolone; CN VII via subtemporal spread; water-exposure trigger; acquired risk factors
36897495	Narrative review	EAC→skull-base anatomical progression
38499694	Cohort (n=39)	Demographics (74% male, 72% T2DM, 75.2 y, PA 88%); diagnostic delay drives extension
32852248	Cohort (n=139)	Facial paralysis 32.4%
23558287	Asian cohort	Clival involvement→persistent disease (p=.002); 63.2% resolution; 21.1% mortality
12737420	Review	Mortality 50%→10–20% with aggressive therapy
40759786	Prospective multicentre SBO (n=235)	PA 57.75%; 16.2% mortality; antibiotic sensitivity profile
35000633	HES analysis	1142% incidence rise 2002–2017
40734530	Updated HES 2002–2024	Rise correlates with ageing + diabetes, not resistance
27241559	Spain epidemiology	Incidence $1.30/10^6$/yr
15807974	Imaging study	CT/MRI + SPECT initial; gallium for follow-up
41287134	PET-CT stewardship	Shorter courses (8 vs 40 wk) without more relapse
40891559	FDG-PET response	PET tracks therapeutic response; ceftazidime+cipro 6.9 wk
27311346	Cohort (n=88)	Prognostic predictors; 5-yr survival 38%; symptom frequencies
30149824	Comparative	Parallel to diabetic foot osteomyelitis
34617899	Case report	Intracranial/vascular spread (Lemierre's)
39534824	Systematic review	Fungal MOE as minority etiology
39828789	Systematic review (surgery)	Surgery reserved for refractory/complicated disease

PMID	Role	Key contribution
36425541	Fungal NOE series	Antifungal management (voriconazole)
42034159	Canine meta-analysis	Canine Pseudomonas OE burden
40937148	HBOT systematic review	HBOT promising, adjunctive for refractory NOE
32209516	Cohort (n=30)	Facial palsy→recurrence (80%, p=.005)
35546515	Surgical series	Facial-nerve surgery best within 14 days

Note on citation integrity: Two treatment citations (PMIDs 39828789 and 36425541) were flagged as snippet "mismatch" during automated validation; the paraphrased claims are consistent with the papers' abstracts/titles as recorded, but the exact verbatim snippet did not match the stored abstract text and should be re-verified against source before final database ingestion.

Limitations and Knowledge Gaps

1. **No molecular/omics data specific to MOE.** No GEO/PRIDE/MetaboLights datasets of MOE tissue exist; there is no transcriptomic, proteomic, metabolomic, lipidomic, single-cell, or spatial characterization of the disease. Diagnosis/monitoring relies on ESR/CRP and imaging, not molecular biomarkers.
2. **No dedicated animal model** of the necrotizing skull-base phenotype; the closest natural correlate (canine Pseudomonas otitis externa) is superficial only.
3. **Mechanism partly inferred.** Several causal-chain steps (host-permissiveness, virulence-factor-driven osteolysis) are inferred from clinical-anatomic correlation and pathogen biology rather than demonstrated in MOE tissue; the foundational review explicitly notes the pathophysiology is incompletely understood.
4. **Heterogeneous, mostly retrospective evidence.** Most data derive from single-center case series and national administrative datasets; terminology (MOE/NOE/SBO/temporal-bone osteomyelitis) is used inconsistently, complicating pooling.

5. **No consensus diagnostic criteria or standardized cure definition** — imaging changes persist after clinical resolution; "cure" and optimal therapy duration remain unstandardized (though FDG-PET is narrowing this gap).
6. **Rising fluoroquinolone resistance** (33–43% in some series) threatens the oral-ciprofloxacin backbone; local antibiograms are essential.
7. **Two citation snippets require re-verification** (see Evidence Base note).

Proposed Follow-up Experiments / Actions

1. **Prospective multicenter registry with standardized diagnostic/cure criteria**, incorporating FDG-PET endpoints, to define optimal therapy duration and validate prognostic scoring (DM + facial palsy + positive CT + age >70).
2. **Molecular characterization of MOE tissue** — obtain EAC granulation-tissue biopsies for host transcriptomics/proteomics and pathogen genomics/biofilm profiling to fill the omics gap and search for prognostic biomarkers.
3. **Develop a diabetic temporal-bone osteomyelitis model** (e.g., diabetic rodent EAC/temporal-bone *P. aeruginosa* biofilm inoculation) to enable mechanistic and preclinical therapeutic studies of the invasive phenotype.
4. **Antimicrobial stewardship trials** testing FDG-PET-guided shortened courses against fixed 6–8-week regimens in a randomized design, with resistance surveillance built in.
5. **Evaluate anti-virulence adjuncts** (quorum-sensing inhibitors, anti-Psl/PcrV antibodies) borrowed from pulmonary Pseudomonas research as add-ons for refractory/multidrug-resistant MOE.
6. **Health-system intervention** targeting early recognition in primary care for elderly diabetics with refractory otalgia, aiming to shorten diagnostic delay (shown to drive disease extension).
7. **Re-verify flagged citations** (PMIDs 39828789, 36425541) against source abstracts prior to knowledge-base ingestion.

End of report.